Contents

Chapter 3

Result And Discussion

Following the experimental and analysis procedures explained in Chapter 2, the ODT capture efficiency was characterized and optimized. This chapter presents the corresponding experimental results.

The first stage of the optimization process consist of minimizing the molasses temperature through the adjustment of the cooling laser detuning, the AOM amplitude, and the compensation coil currents.

Subsequently, the ODT loading is optimized by exploiting the compensation coils to displace the MOT after loading. This optimization aims to maximize the spatial overlap between the atomic cloud and the ODT beam, thereby enhancing the capture efficiency.

3.1 Molasses temperature optimization

Before performing any optimization, the six cooling beams were realigned to minimize their overlap mismatch at the MOT position. This procedure is crucial since thermal drifts can induce misalignments in the optical components over time.

Preliminary values for the compensation coil currents were obtained prior to the realignment using the Release-and-Recapture (R&R) technique:

$$\begin{aligned} I_{\mathrm{x}} &= 80\ \mathrm{mA}, \\ I_{\mathrm{y}} &= -200\ \mathrm{mA}, \\ I_{\mathrm{z}} &= -400\ \mathrm{mA}. \end{aligned} \tag{3.1}$$

These values were used as the starting point for the subsequent optimization procedure.

3.1.1 Detuning frequency optimization

The first optimization step consisted of measuring the molasses temperature for different values of the cooling laser detuning. The detuning was varied from -24.4 MHz ($4.0\,\Gamma/2\pi$)

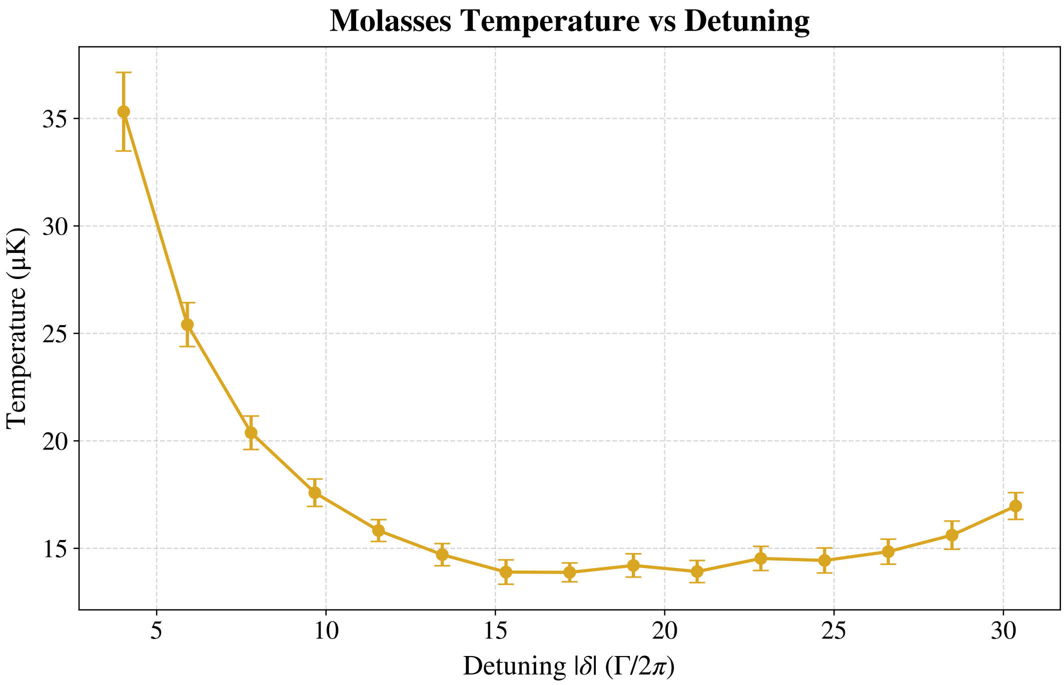


Figure 3.1: *Optical molasses temperature as a function of laser detuning. The horizontal axis represents the absolute value of the applied red detuning* $|\delta|$*, normalized to the natural linewidth* $\Gamma/2\pi$*. The data exhibits optimal sub-Doppler cooling in the region* $|\delta| \approx 15-20$*, reaching a minimum temperature of* 13.9 ± 0.4 μK*. For detunings up to* $|\delta| \approx 15$*, the temperature follows the* $1/|\delta|$ *dependence expected for an ideal Sisyphus cooling. At larger detunings, the curve flattens and eventually begins to rise.*

to -184 MHz ($30.4\ \Gamma/2\pi$), with a step size of -11.4 MHz.

The resulting average temperature is shown in Figure 3.1 as a function of the detuning expressed in units of $\Gamma/2\pi$. At small detunings, the data follow the expected $1/|\delta|$ dependence typical of the Sisyphus cooling. As the detuning increases, two phenomena become relevant: a significant drop in the optical scattering rate $\Gamma_p \propto 1/\delta^2$ and the approach of the transition $F = 2 \to F' = 2$ located 267 MHz below the cooling transition $F = 2 \to F' = 3$ (see Figure 2.2). These effects cause the temperature curve to flatten in the region $|\delta| \approx 15 - 20$ and eventually to rise again for larger detunings.

The lowest average temperature was obtained for a normalized detuning of $|\delta| = 17.2$ corresponding to an absolute detuning of $|\delta| = 104.2$ MHz. Figure 3.2 shows the fit of equation (2.9) for this optimal configuration. The resulting temperatures along the Cartesian axes are $T_x = (14.1 \pm 0.5)$ μK and $T_y = (13.5 \pm 0.7)$ μK. Their weighted average yields a final temperature of $T = (13.9 \pm 0.4)$ μK.

As observed in Figure 3.2, the data points at long TOF diverge appreciably from the linear trend, consistently falling below the expected values. This behavior is systematically observed across all measurements. The hypothesis is that at sufficiently long TOF the cloud radius becomes non-negligible compared to the probe beam radius, violating the assumption of uniform illumination. Consequently, the outer region of the cloud interacts

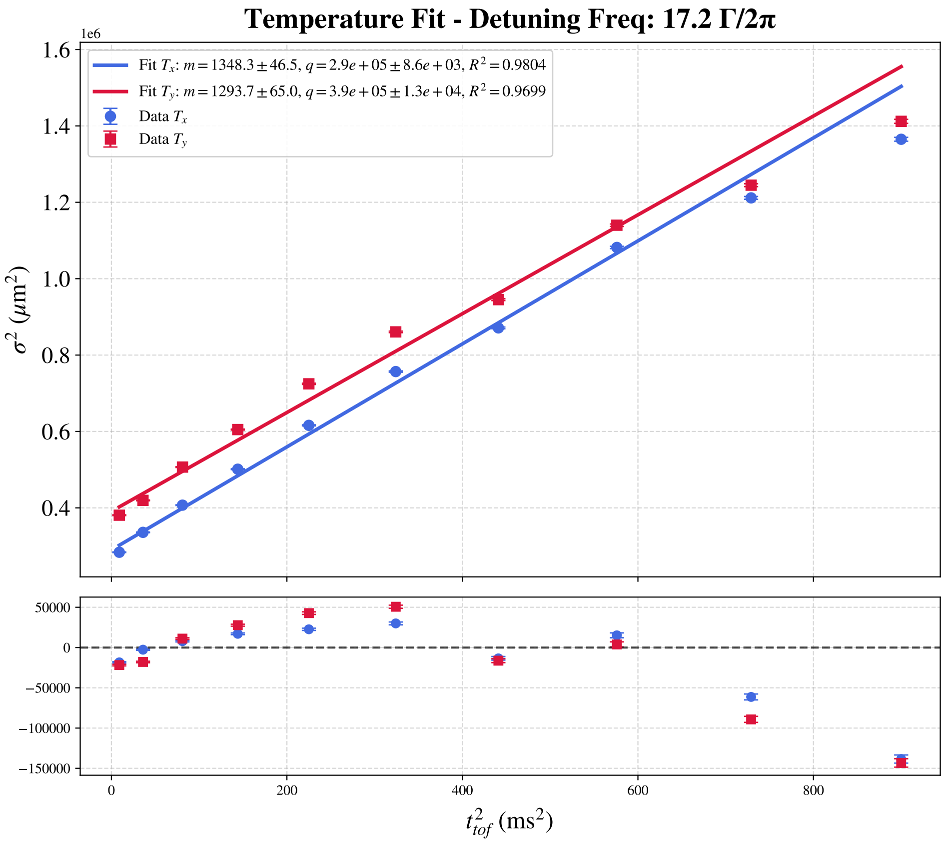


Figure 3.2: *Optical molasses temperature fit for $|\delta| = 17.2\ \Gamma/2\pi$ with residuals. The large residuals of the last three data points from the linear fit are attributed to the cloud expansion, which makes its radial size non-negligible compared to the probe beam profile.*

with a lower intensity field, causing the Gaussian fit to systematically underestimate the true variance. Fortunately, this effect is expected to become less prominent as the temperature is lowered.

3.1.2 AOM amplitude optimization

As a second optimization step, the influence of the cooling beam power was investigated by varying the AOM drive amplitude and the duration of its transition ramp, which is applied at the beginning of the molasses phase. The AOM amplitude, expressed in arbitrary units (a.u.), controls the RF amplitude driving the AOM. Three values were tested: 385, 475 and 550 a.u., corresponding respectively to approximately 0.7, 0.85 and 1.0 times the AOM amplitude during the MOT phase. Additionally, ramp durations of 5 ms and 20 ms were tested for the transition from the MOT value to the molasses value. These measurements were repeated for different values of the cooling detuning. The optimization results are shown in Figure 3.3. Although small variations in the expected values of the temperature can be observed for different values of the AOM amplitude, these differences fall within the experimental error bars. Therefore, no statistically significant dependence of the molasses

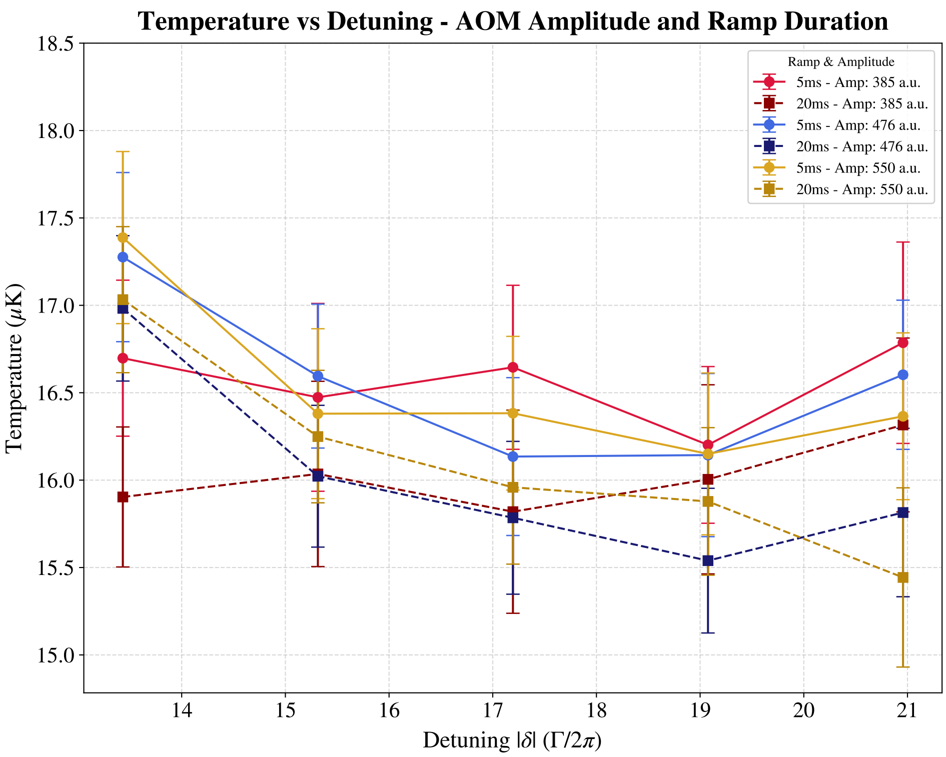


Figure 3.3: *Optical molasses temperature versus cooling detuning for different AOM amplitude and ramp durations. No clear optimal AOM amplitude is observed as the variations remain within the uncertainty. Temperatures corresponding to a ramp duration of 20 ms (dashed lines) are systematically lower than those obtained with a 5 ms ramp (solid lines).*

temperature on the AOM amplitude was found, and no clear optimum could be identified within the tested range. Lower amplitudes were not investigated to avoid operating the AOM too far from its optimal efficiency regime.

Conversely, as shown in Figure 3.3, the temperatures obtained with a 20 ms ramp are systematically lower than those obtained with a 5 ms ramp, yielding an average temperature reduction of approximately 0.5 μK. This improvement was relatively small, and was considered sufficient for the scope of the experiment. Since further optimization on the order of a few tenths of a microkelvin was not required, no other ramp times were tested. Consequently, a ramp duration of 20 ms was selected for the subsequent measurements.

It is worth noting that during the ODT loading characterization, this optimization was found to introduce additional fluctuations into the system. For this reason, the subsequent optimization of the compensation coil currents was carried out using an AOM amplitude of 385 a.u. (corresponding to the lowest temperature measured at $|\delta| = 17.2\ \Gamma/2\pi$) with a ramp duration of 20 ms. However, these parameters were later readjusted during the ODT optimization. Nevertheless, the measurements presented in this section revealed that their influence on the molasses temperature is limited to a few tenths of a microkelvin, and therefore does not significantly affect the final achieved temperature.

3.1.3 Compensation field optimization

The final stage of the molasses optimization procedure consisted of adjusting the compensation magnetic field. The compensation coil currents are set to the molasses values 5 ms after the MOT magnetic field had been switched off.

The optimization was performed sequentially. First, the z-axis current was optimized while keeping the x- and y-axes currents fixed to their preliminary values (3.1). Then the x-axis current was optimized using the newly found optimal z-value. Finally, the y-axis current was optimized using the optimal currents for both the z- and x-axes. Quadratic fits were applied to the experimental data to extract the optimal current values and their corresponding minimum temperatures. The results are shown in Figures 3.4, 3.5 and 3.6 for the z-, x-, and y-axis coils respectively.

The resulting optimized currents, used in the subsequent ODT loading stage, are:

$$\begin{aligned} I_{\mathrm{x}} &= -15 \text{ mA}, \\ I_{\mathrm{y}} &= -185 \text{ mA}, \\ I_{\mathrm{z}} &= -230 \text{ mA}. \end{aligned} \tag{3.2}$$

These parameters yielded a minimized molasses temperature of $T = (9.6 \pm 0.1)\ \mu\text{K}$. The z-axis compensation coil current deviated the most from its preliminary value, while the x-axis current underwent a smaller but still appreciable correction. In contrast, the y-axis coil current remained close to its initial value.

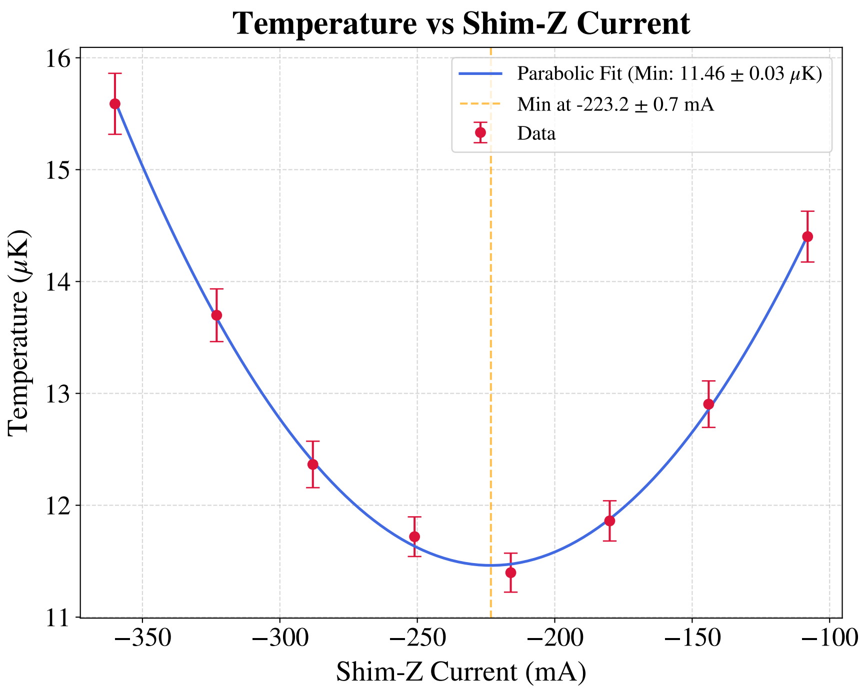


Figure 3.4: *Optical molasses temperature versus Shim-Z current. The parabolic fit yields a minimum temperature* $T = (11.46 \pm 0.03)\ \mu\text{K}$ *at a current* $I_z = (-223.2 \pm 0.7)$ *mA.*

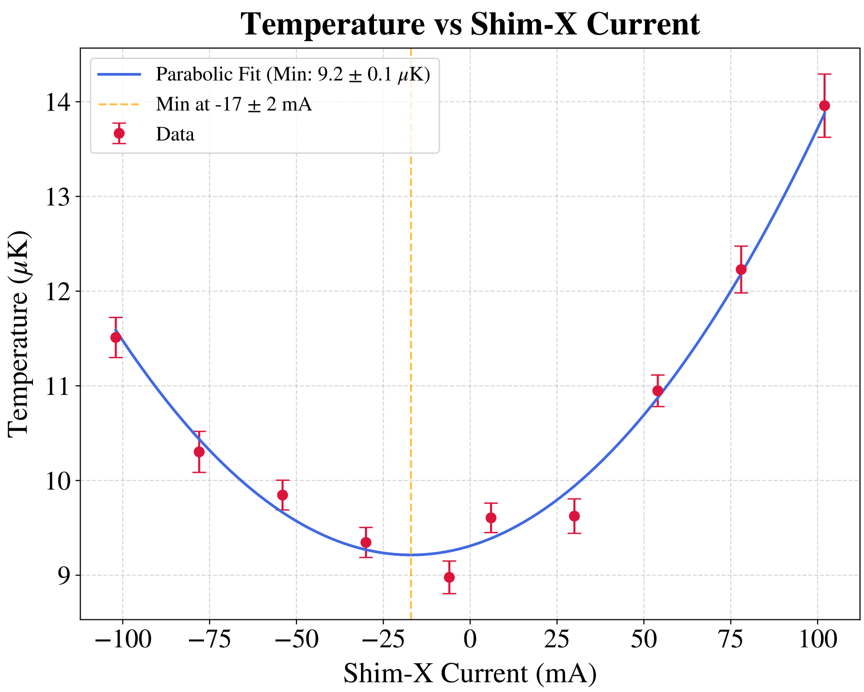


Figure 3.5: *Optical molasses temperature versus Shim-X current. The parabolic fit yields a minimum temperature* $T = (9.2 \pm 0.1)$ μK *at a current* $I_x = (-17 \pm 2)$ *mA.*

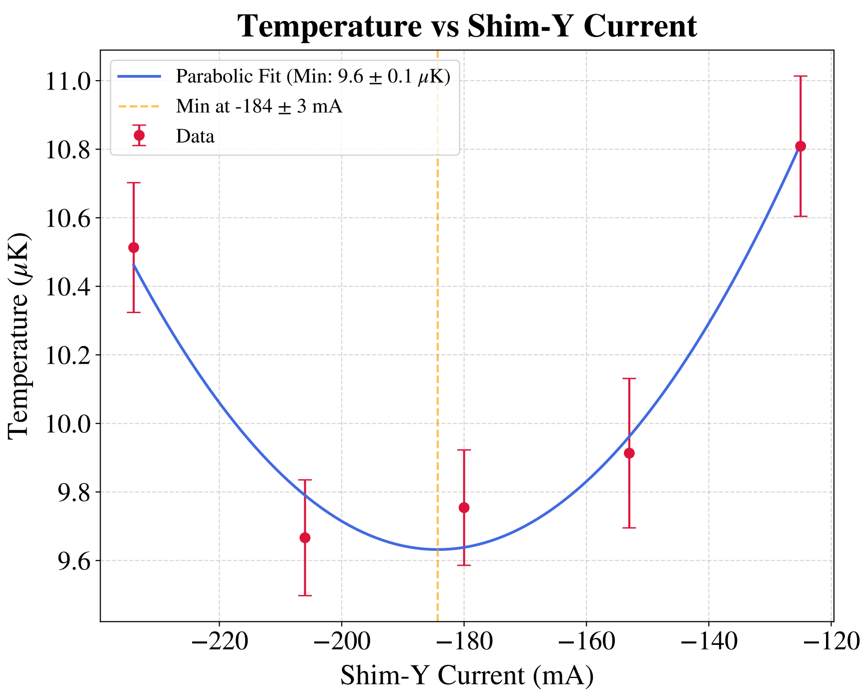


Figure 3.6: *Optical molasses temperature versus Shim-Y current. The parabolic fit yields a minimum temperature of* $T = (9.6 \pm 0.1)$ μK *at a current* $I_y = (-184 \pm 3)$ *mA.*

3.2 ODT loading optimization

The optimal parameters obtained for the molasses were used for the optimization of the ODT loading. This procedure primarily focused on tuning the compensation magnetic field. The underlying concept is to first load the MOT and then, prior to the molasses phase, exploit the compensation coils to spatially shift the MOT position.

Because the MOT forms at the zero of the magnetic quadrupole field, adjusting the compensation coil currents effectively translates the field zero, displacing the trapped atomic cloud. Moving the MOT away from the optimal intersection volume of the beams inevitably causes some atom loss and a temperature increase. However, this is a necessary trade-off to achieve a better spatial overlap with the ODT beam, which ultimately maximizes the number of atoms transferred into the dipole trap. As long as this spatial shift is small, the optimal molasses parameters derived in the previous sections remain effective at nullifying the residual magnetic field, requiring no further adjustments

Specifically, after the initial MOT loading phase, the compensation coil currents are ramped to their ODT loading values over a duration of 200 ms, discretized in 20 steps. This smooth translation is crucial to minimize atom loss and heating during the transport process.

Similarly to the molasses phase, preliminary current values for the ODT loading, obtained prior to the optical realignment and the molasses optimization, were available as a starting point:

$$\begin{aligned} I_\mathrm{x} &= -200 \text{ mA} \\ I_\mathrm{y} &= -350 \text{ mA} \\ I_\mathrm{z} &= 50 \text{ mA}. \end{aligned} \tag{3.3}$$

3.2.1 Imaging system optimization

Initial measurements were performed using the FLIR camera, employing the optimal parameters derived from the molasses temperature optimization and the preliminary compensation coil currents for the ODT loading. The data were analyzed to characterize the ODT capture efficiency using a procedure analogous to that described in Section 2.3.3, though without background rescaling. The resulting intensity difference images are shown in Figure 3.7 (2D image) and Figure 3.8 (1D integrated profile). Visual inspection of the 2D image does not reveal any distinct brighter region attributable to the atoms captured by the ODT. Conversely, the 1D integrated profile shows a localized intensity increase spanning ~ 4 pixels around the fiber position. Subsequent measurements revealed a similar behavior, confirming that this localized peak corresponds to the atoms trapped in the ODT. However, the signal amplitude was often too small to be reliably distinguished from the noise. Since the 1D integration is explicitly intended to enhance the signal-to-noise

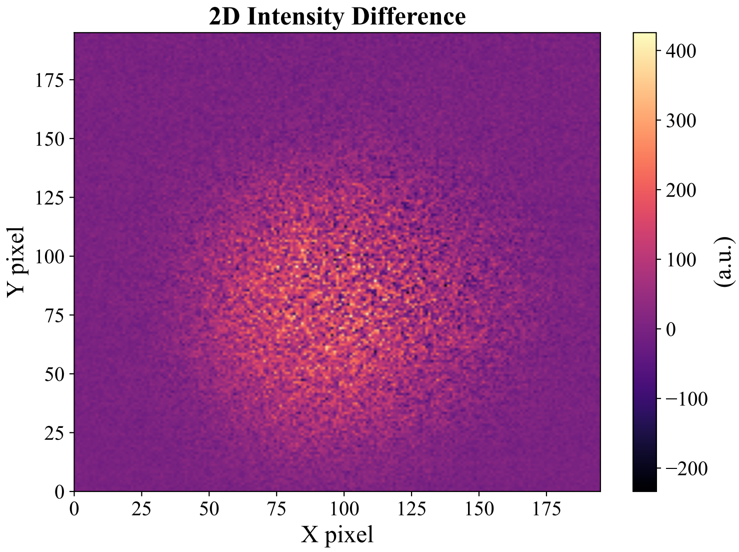


Figure 3.7: *Intensity difference 2D image, obtained with the FLIR camera and a 20 ms AOM amplitude ramp duration during the molasses phase. The intensity distribution does not show any brighter area that can be attributed to the presence of the ODT.*

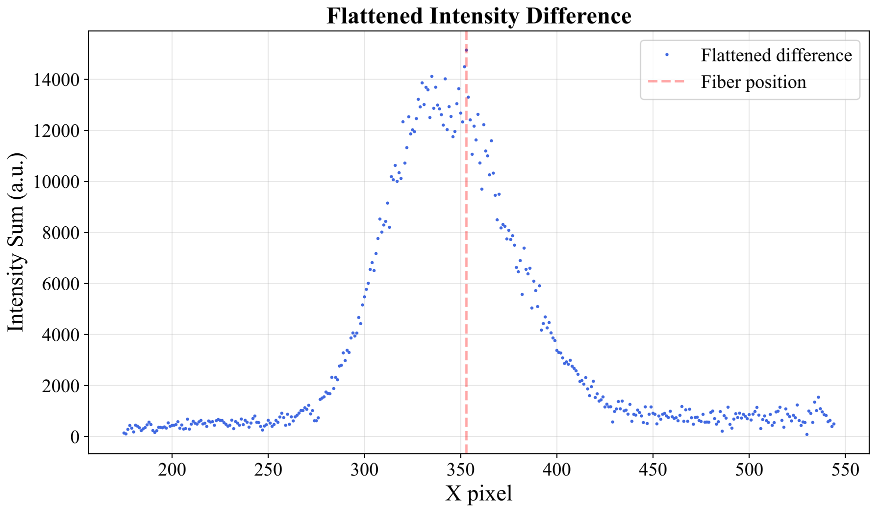


Figure 3.8: *Intensity difference 1D integrated profile, obtained with the FLIR camera and a 20 ms AOM amplitude ramp during the molasses phase. The red vertical line indicates the approximate position of the fiber center. A few higher-intensity pixels are recognizable corresponding to the ODT position, but the averaging and subtracting procedures are not effective in removing the background.*

ratio (SNR), the persistence of a poor SNR highlighted a critical issue. Furthermore, the presence of a residual background pedestal significantly larger than the ODT signal indicates underlying oscillatory noise in the system that is not averaged by the acquisition of five images, suggesting either a large standard deviation or a correlated shot-to-shot drift. Beyond the noise, the magnitude of the ODT signal itself was a primary concern. Even far from an optimal atomic-cloud-ODT overlap, a higher initial capture was expected. Additionally, the limited spatial extent of the ODT signal (occupying only a few pixels)

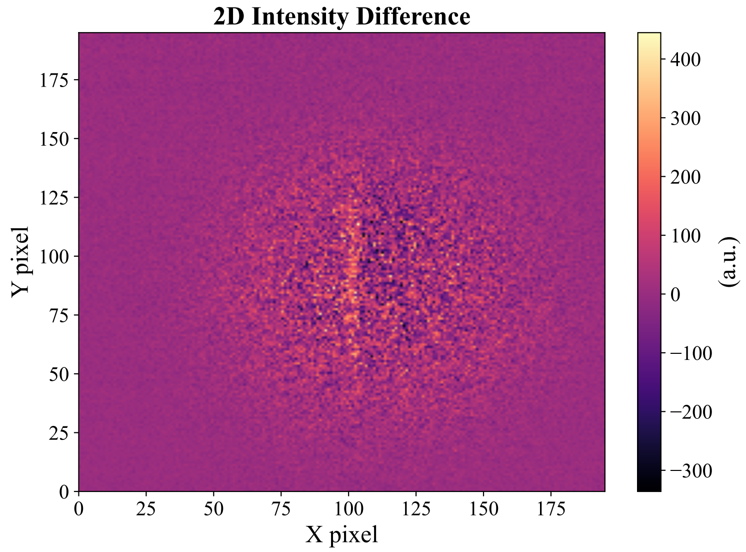


Figure 3.9: *Intensity difference 2D image, obtained with the FLIR camera and a 5 ms AOM amplitude ramp duration during the molasses phase. A brighter area corresponding to the ODT is clearly visible.*

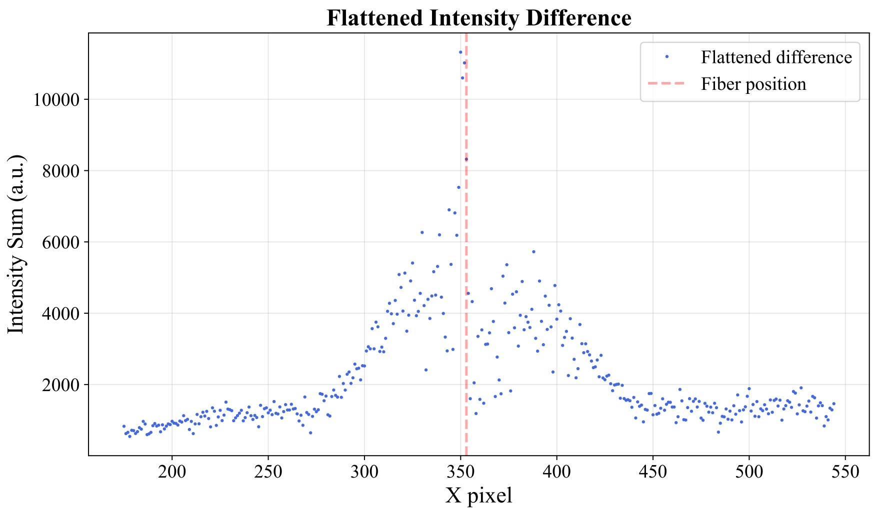


Figure 3.10: *Intensity difference 1D integrated profile, obtained with the FLIR camera and a 5 ms AOM amplitude ramp during the molasses phase. The red vertical line indicates the approximate position of the fiber center. With the 5 ms ramp, the averaging and the subtraction become more effective in separating the ODT signal from the background. However, the problem of the small number of pixels representing the ODT signal is still present.*

posed a severe analytical challenge. Such poor spatial resolution precludes reliable 2D Gaussian fitting, and extracting the signal area via direct pixel summation introduces systematic errors. In summary, the diagnostic capabilities were limited by two main factors: first, the residual background in the subtracted images overwhelmed the weak ODT signal; second, the small waist of the ODT beam resulted in an inadequate number of illuminated pixels for robust analysis.

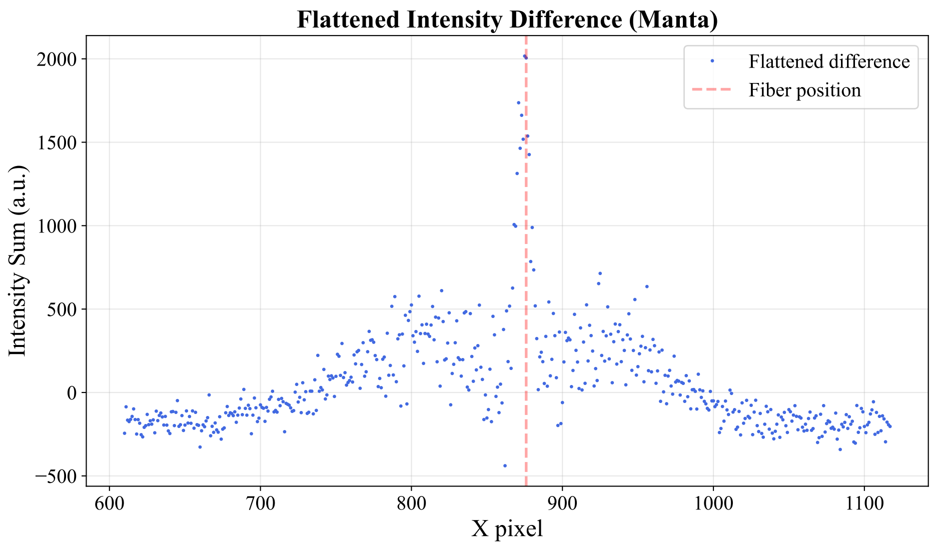


Figure 3.11: *Intensity difference 1D integrated profile, obtained with the Manta camera and a 5 ms AOM amplitude ramp during the molasses phase. The combination of the Manta's smaller pixel size and the closer mounting position effectively increased the spatial resolution of the ODT signal.*

To address the poor SNR, the molasses parameters were systematically investigated. It was found that the noise was strongly dependent on the AOM amplitude ramp during the molasses phase. In particular, reducing the ramp duration from 20 ms to 5 ms significantly improved the background subtraction and increased the SNR. Furthermore, to minimize intensity fluctuations induced by the AOM driver, the target amplitude was set to 480 a.u. This value corresponds to the saturation regime of the AOM diffraction efficiency, which acts as a working point with a minimum derivative ($dP/dV \approx 0$). Operating in this saturation region intrinsically suppresses amplitude noise, thereby stabilizing the actual optical power delivered during the molasses phase.

The results obtained with this optimized configuration are shown in Figures 3.9 (2D difference) and 3.10 (1D integrated profile). In the 2D plot, a distinct brighter strip representing the atoms captured in the ODT is now recognizable. By comparing the 1D integrated profiles in Figures 3.8 and 3.10, it is evident that the pedestal in the 5 ms ramp case is suppressed, showing a maximum amplitude and area approximately half those of the 20 ms ramp case. This reduction is sufficient to make the ODT signal emerge clearly from the residual pedestal and noise.

Despite the improved SNR, a critical issue remained: the insufficient spatial resolution provided by the FLIR camera setup. To overcome this limitation, subsequent measurements were performed using the Manta camera, which features a smaller pixel pitch. Additionally, the Manta camera was mounted closer to the experimental chamber to exploit the minimum working distance of the MVL50M23 objective, maximizing the optical magnification. Comparing the apparent fiber width measured by both cameras, it was deduced that the object size in pixels increased by a factor of ~ 2.25 with the Manta setup. This scaling factor arises from the combined contribution of the higher optical magnification and the smaller pixel size, finally yielding a sufficiently resolved ODT signal for robust analysis, as shown in Figure 3.11.

3.2.2 Capture efficiency optimization

The systematic optimization of the compensation coil currents was carried out following the complete analysis procedure detailed in Section 2.3.3, utilizing the Manta camera. The procedure was performed sequentially along the spatial axes in the order $x \to z \to y$. During each scan, the currents of the remaining non-optimized coils were maintained at the preliminary values established in Equation (3.3).

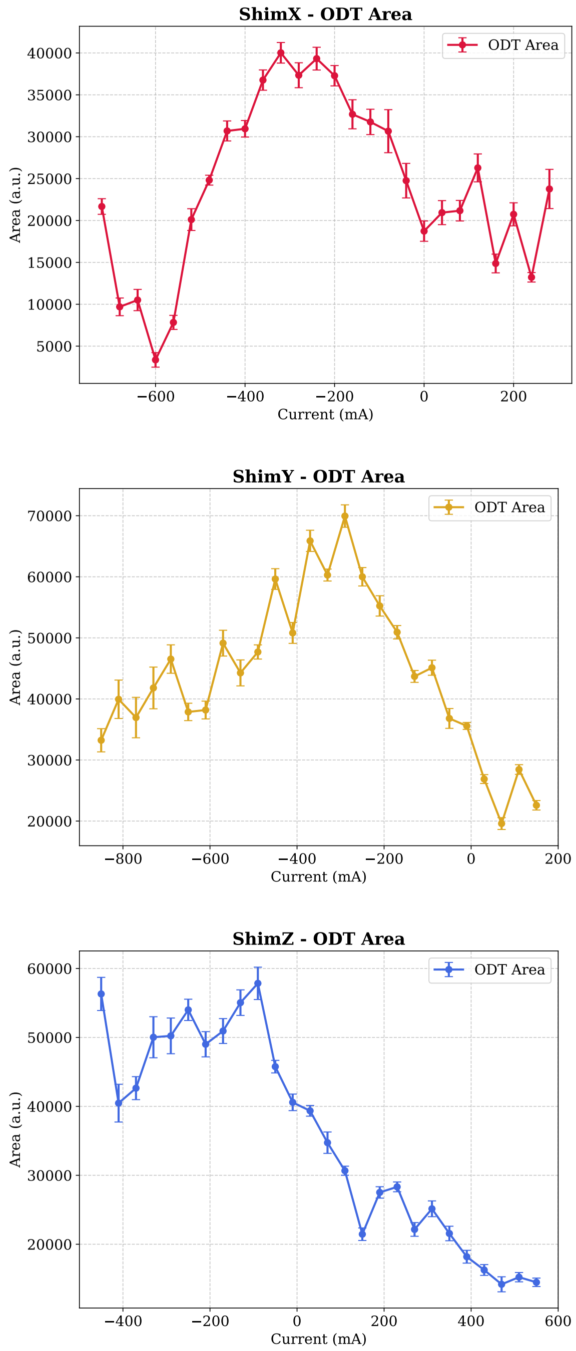
The capture efficiencies extracted from these scans are summarized in Figure 3.12b for the x-, y-, and z-coils. Similarly, Figure 3.12a displays the evaluated ODT signal area for each scan. This area is directly proportional to the total number of captured atoms, making it the physical quantity of fundamental interest. However, since the camera was not calibrated for absolute atom number measurements, this quantity can only be expressed in arbitrary units. Nonetheless, these values remain perfectly comparable provided the camera gain is kept constant. As evident from the plots, the behavior of the capture efficiency mirrors the trend of the ODT area.

This initial optimization yielded the following optimal currents:

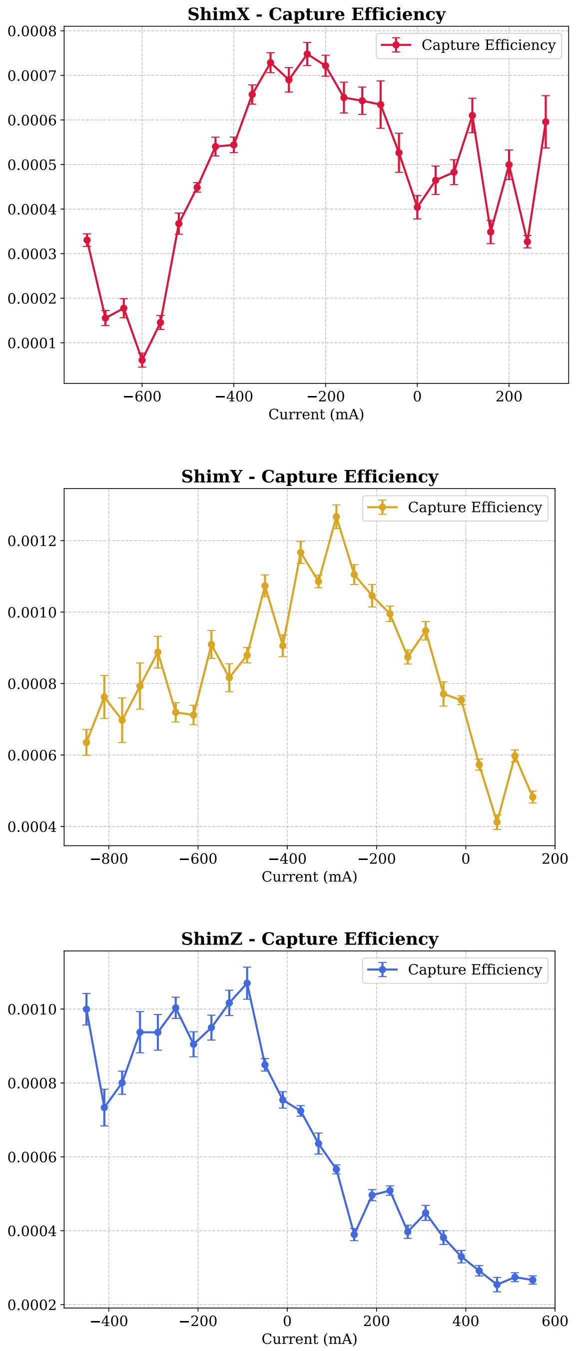
$$
\begin{aligned}
I_\mathrm{x} &= -220 \text{ mA}, \\
I_\mathrm{y} &= -290 \text{ mA}, \\
I_\mathrm{z} &= -90 \text{ mA}.
\end{aligned}
\tag{3.4}
$$

To confirm the optimization achieved in the initial measurements, a second run was performed, yielding unconventional results regarding the z-axis compensation coil (Figure 3.13). As clearly shown in Figure 3.13a, the signal does not follow an expected parabolic or bell-shaped profile with a clear maximum. Instead, the ODT area and the capture efficiency exhibit an oscillatory behavior. Starting from a current of -1000 mA, the signal increases to a first local maximum at -800 mA, drops until ~ -700 mA, and then rises again to reach subsequent local maxima at ~ -600 mA and ~ -400 mA.

A similar periodic trend is observed in the MOT background area which represents the atomic cloud area measured at the end of the TOF with the ODT turned off (Figure 3.13b). This area exhibits peaks at ~ -800 mA, ~ -600 mA, and ~ -400 mA, matching the ODT area oscillation with good precision. While a proportional increase in the captured ODT fraction is naturally expected when the initial MOT atom number is higher, the oscillatory nature of this trend is peculiar. Tracking the x-centroid of the atomic cloud



(a) *ODT area.*



(b) *Capture efficiency.*

Figure 3.12: *ODT area (left) and capture efficiency (right) for the compensation coil current scans along x-, y-, and z-axes (performed in the order $x \rightarrow z \rightarrow y$). The qualitative agreement between the two observables is clearly visible for all the investigated axes.*

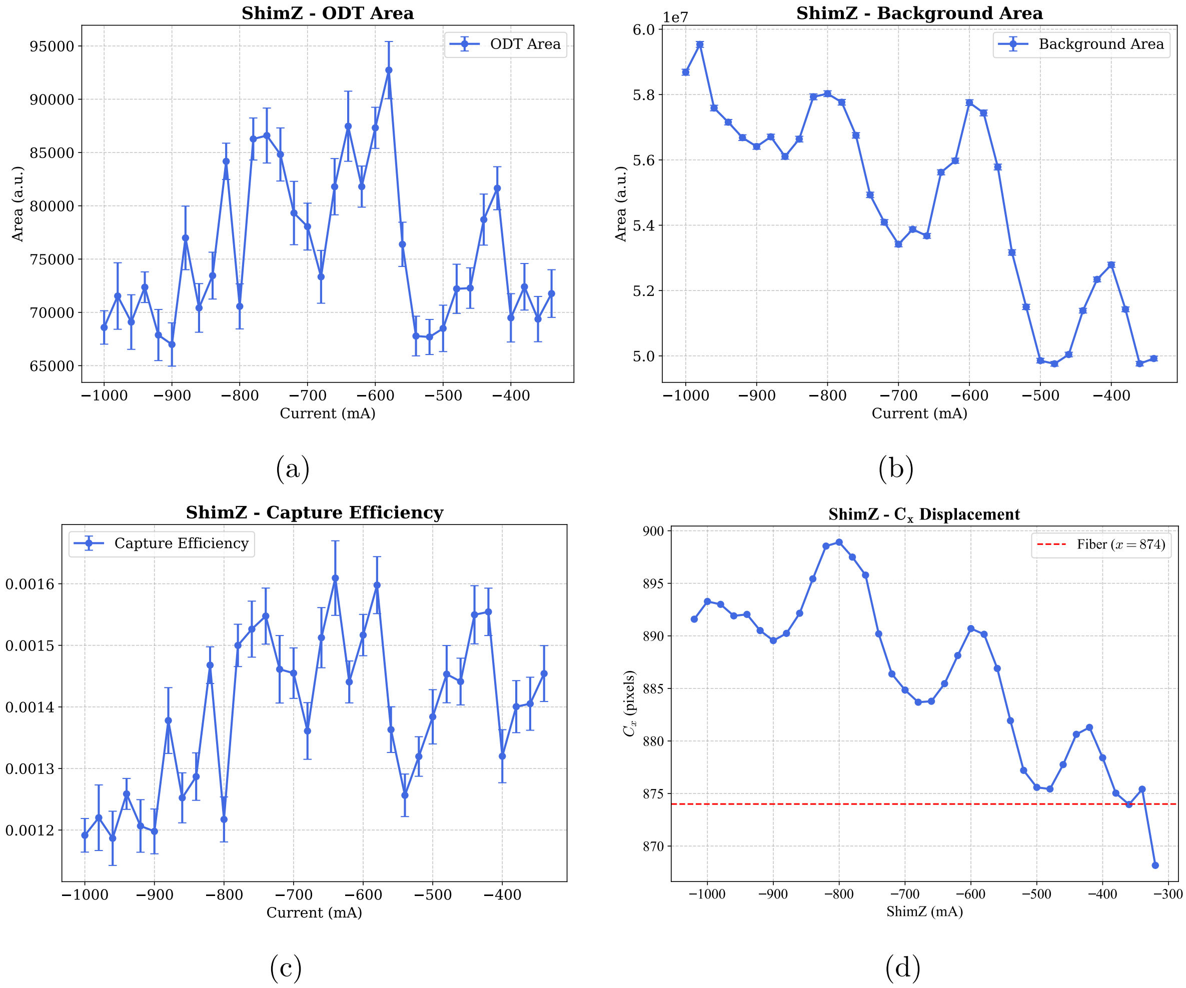


Figure 3.13: *Optimization results for the second set of measurements of the z-compensation coil current: (a) ODT signal area, (b) background area, (c) capture efficiency, (d) atomic cloud center position projected on the camera's x-axis.*

on the camera's image plane (Figure 3.13d) reveals an analogous spatial oscillation, a phenomenon not observed during the scans of the other two compensation axes. The likely physical explanation for this behavior lies in an imperfect spatial overlap of the cooling beams. As the z-axis compensation field is scanned, the position of the magnetic zero shifts along the laboratory z-axis as expected. However, a non-ideal beam intersection generates a complex 3D optical intensity landscape with local gradients. Since the MOT equilibrium position is determined by the local balance of the radiation pressures, the atomic cloud dynamically adapts its position to the local intensity imbalances. Instead of a linear translation along z, the cloud is pushed along a trajectory that locally nullifies the net optical force, resulting in a combined motion in the laboratory z–x plane that projects as a spatial oscillation onto the 2D camera sensor. Furthermore, as the MOT position change in this complex optical landscape, it encounters regions of varying laser intensity. As a consequence, both the cooling efficiency and the trap depth fluctuate accordingly. Consequently, the atom loss and the final MOT temperature vary depending on the exact spatial position of the cloud. This spatial dependence explains the observed fluctuations in the background are, and directly impacts the subsequent capture efficiency into the ODT.

Finally, the optimal configuration maximizing the ODT loading was found to be:

$$\begin{aligned} I_{\mathrm{x}} &= -220 \text{ mA}, \\ I_{\mathrm{y}} &= -280 \text{ mA}, \\ I_{\mathrm{z}} &= -600 \text{ mA}. \end{aligned} \tag{3.5}$$

This configuration yielded the largest ODT trap area, representing the maximum number of captured atoms, and is associated with a capture efficiency of $(0.160 \pm 0.005)\%$. The corresponding 2D difference image is shown in Figure 3.14.

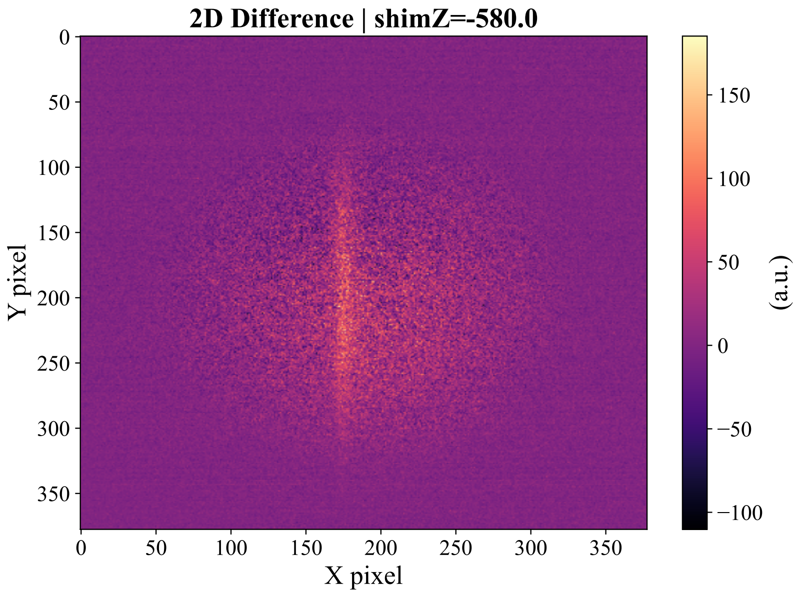


Figure 3.14: *Intensity difference 2D image obtained with the optimal currents:* $I_{\mathrm{x}} = -220$ mA, $I_{\mathrm{y}} = -280$ mA, *and* $I_{\mathrm{z}} = -600$ mA, *corresponding to a capture efficiency of* $\sim 0.16\%$.